

Answer Key — Particulate Nature of Matter

Final answers with brief reasons.

Objective

Q1 (a) **constituent particles** — the basic units are constituent particles.

Q2 (b) **liquid** — liquids: definite volume, no fixed shape.

Q3 (c) **solid** — solids have the strongest forces.

Q4 (b) **gases** — gases diffuse fastest.

Q5 (a) **gas** — gases compress easily due to large spaces.

Q6 (b) **False** — it is a physical change — no new substance forms.

Q7 **interparticle** — these are interparticle spaces.

Q8 **melting** — that is the melting point.

Short answer

Q9 Solid: definite shape and volume; Liquid: definite volume, no fixed shape; Gas: no fixed shape or volume.

Q10 Gases have large interparticle spaces, so particles can be pushed closer; solids/liquids have little space, so they are almost incompressible.

Q11 The spreading of particles of one substance into another on their own; fastest in gases.

Long / application

Q12 Heating makes particles vibrate more; at the melting point they break free of fixed positions and forces weaken — solid → liquid. Further heating makes particles move fast enough to escape; forces become negligible — liquid → gas (at the boiling point).

Q13 (i) Sugar splits into tiny particles that fit into the interparticle spaces between water particles, so little extra volume is added. (ii) Incense particles diffuse through the air; gas particles move fast with large spaces, so the smell spreads quickly.
